

Problem

Solve Prob. 5.19 using *PSpice* or *MultiSim* and op amp uA741.

Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

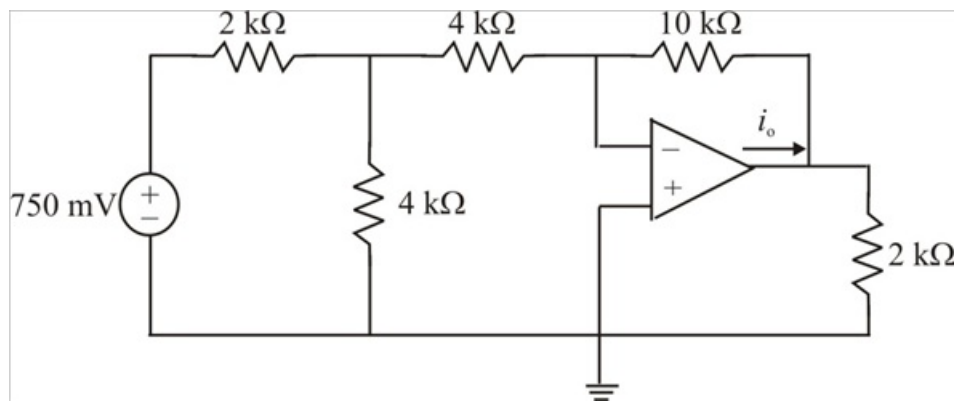


Figure 1

Step 2 of 6

Label the circuit as shown in Figure 2:

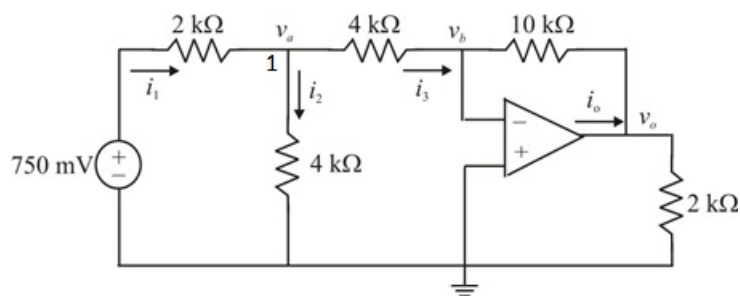


Figure 2

Step 3 of 6

Apply Kirchhoff's current law to node 1 of circuit in Figure 2.

$$i_1 = i_2 + i_3$$

$$\frac{750 \text{ mV} - v_a}{2 \text{ k}\Omega} = \frac{v_a - 0}{4 \text{ k}\Omega} + \frac{v_a - v_b}{4 \text{ k}\Omega}$$

$$\frac{750 \times 10^{-3} \text{ V} - v_a}{2 \times 10^3 \Omega} = \frac{v_a + v_a - v_b}{4 \times 10^3 \Omega}$$

$$1.5 - 2v_a = 2v_a - v_b$$

$$4v_a - v_b = 1.5 \dots\dots (1)$$

But the voltage $v_b = 0$, since the non-inverting terminal of the operational amplifier is grounded.

Substitute 0 for v_b in equation (1).

$$4v_a - v_b = 1.5$$

$$4v_a - 0 = 1.5$$

$$v_a = 0.375 \text{ V}$$

Step 4 of 6

Apply Kirchhoff's current law to node 2 of circuit in Figure 2.

$$i_3 = i_o$$

$$\frac{v_a - v_b}{4 \text{ k}\Omega} = \frac{v_b - v_o}{10 \text{ k}\Omega}$$

$$5v_a - 5v_b = 2v_b - 2v_o$$

$$2v_o = 7v_b - 5v_a \dots\dots (2)$$

Substitute 0.375 V for v_a , and 0 for v_b in equation (2).

$$\begin{aligned} 2v_o &= 7v_b - 5v_a \\ &= 7(0) - 5(0.375 \text{ V}) \\ &= -1.875 \text{ V} \end{aligned}$$

$$\begin{aligned} v_o &= \frac{-1.875 \text{ V}}{2} \\ &= -0.9375 \text{ V} \end{aligned}$$

Step 5 of 6

Apply Kirchhoff's current law at output node v_o .

$$i_o = \frac{v_o - v_b}{10 \text{ k}\Omega} + \frac{v_o - 0}{2 \text{ k}\Omega}$$

Substitute 0 for v_b .

$$\begin{aligned} i_o &= \frac{v_o}{10 \text{ k}\Omega} + \frac{v_o}{2 \text{ k}\Omega} \\ &= \frac{v_o + 5v_o}{10 \text{ k}\Omega} \\ &= \frac{6v_o}{10 \text{ k}\Omega} \end{aligned}$$

Substitute -0.9375 V for v_o .

$$\begin{aligned} i_o &= \frac{6(-0.9375 \text{ V})}{10 \times 10^3 \Omega} \\ &= \frac{-5.625 \text{ V}}{10 \times 10^3 \Omega} \\ &= -562.5 \times 10^{-6} \text{ A} \\ &= -562.5 \mu\text{A} \end{aligned}$$

Therefore, the current i_o is $-562.5 \mu\text{A}$.

PSpice Analysis:

Draw the circuit as shown in Figure 3 using the Schematics:

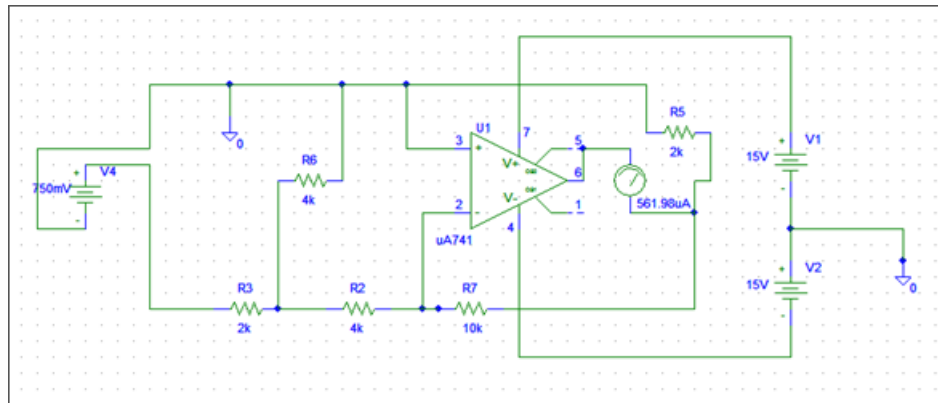


Figure 3

Step 6 of 6

Notice that the positive terminal of the voltage source is connected to the inverting terminal (pin 2) via resistors, while non-inverting terminal is connected to pin 3, and also notice that the positive power supply terminal (pin 7) is connected to a 15 V dc voltage source, while the negative power supply terminal (pin 4) is connected to -15 V . The pins 1 and 5 are left floating because they are used for offset null adjustment. Add IPROBE to measure the output current of op amp.

We have an almost identical answer using the computer to analyze the circuit. We can see the output current, IPROBE reads $561.98 \mu\text{A}$ and compares this to $562.5 \mu\text{A}$ of theoretical value.

A simplified analysis has produced the nearest result as a computer analysis as shown in Figure 3.